

TrendChecker: Deployment, Setup & Operational User Manual

Operations, Deployment & User Handbook

Project: TrendChecker (SSM)

Target Audience: Systems Administrators, Research Faculty, Deployment Engineers

Version: 1.0.0 (Production Release)

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1. System Requirements & Prerequisites

TrendChecker is engineered for lightweight, cross-platform operation without requiring dedicated GPU acceleration or third-party cloud AI tokens.

1.1 Minimum & Recommended Hardware Specifications

Hardware Component	Minimum Requirement	Recommended Specification
Processor (CPU)	64-bit Dual-Core (2.0 GHz)	64-bit Quad-Core (Intel i5/i7, AMD Ryzen, Apple M-Series)
System Memory (RAM)	4 GB Available RAM	8 GB – 16 GB High-Speed RAM
Disk Storage	250 MB free disk space	1 GB (allowing for persistent SQLite historical logs)
Network Interface	Standard Broadband Connection	High-Speed Low-Latency Internet
Graphics (GPU)	None required (GPU-Free)	Integrated or Discrete (CPU vector acceleration)

1.2 Supported Operating Systems

- **Microsoft Windows:** Windows 10, Windows 11, Windows Server 2019/2022 (x86_64).
- **Linux:** Ubuntu 20.04+, Debian 11+, RHEL/CentOS 8+, Arch Linux (x86_64 / aarch64).
- **Apple macOS:** macOS Monterey (12.0)+, macOS Ventura, macOS Sonoma, macOS Sequoia (Intel & Apple Silicon M1-M4).

1.3 Python Runtime Environment

- **Supported Versions:** Python 3.10, 3.11, 3.12, 3.13.
- **Verified Runtime:** Python 3.13.5 (64-bit).

2. Step-by-Step Installation & Setup

Follow these four steps to set up and verify the TrendChecker environment from scratch:

1 Open Terminal / PowerShell in Project Root

Navigate to the root directory containing `run_web.py` and `requirements.txt`:

```
cd "d:\AiTeC Internship '26\TrendChecker - SSM\social-media-monitoring"
```

2 Create a Clean Python Virtual Environment (Recommended)

Isolating dependencies ensures zero conflicts with system-level packages:

```
# Windows PowerShell
python -m venv venv
. env\Scripts\Activate.ps1

# Linux / macOS Bash
python3 -m venv venv
source venv/bin/activate
```

3 Install Production Dependencies

Install all required web, forensic, and scraping libraries:

```
pip install --upgrade pip
pip install -r requirements.txt
```

4 Verify Installation with Automated Test Suite

Execute the built-in test suite to verify that all collectors, models, and forensic engines are functioning properly:

```
python -m pytest tests/ -v
```

Expected Output: `38 passed in ~10s (100% pass rate).`

3. System Execution & Runtime Interfaces

TrendChecker provides two independent runtime interfaces: an interactive **Cyber-Glass Web Dashboard** and a scriptable **Terminal Command-Line Interface (CLI)**.

Execution Mode	Startup Command	Primary Use Case
Mode A: Web Dashboard Primary Interface	<code>python run_web.py</code>	High-performance asynchronous FastAPI web server on <code>http://127.0.0.1:8000</code> . Full interactive dashboard with live video/image verification.
Mode B: Terminal CLI Headless / Batch	<code>python run_cli.py</code>	Scriptable command-line interface for querying specific platforms, setting post retrieval limits, inspecting JSON structures, and batch logging.

4. Web Dashboard User Manual

The web dashboard features a modern **Cyber-Glass UI** designed for high-density intelligence monitoring:

Cyber-Glass Dashboard UI Layout Overview

```
+-----+
| [⚡ 7-SOURCES] TrendChecker      (● System Active)      [⚙️ Scraper Settings] |
+-----+
| [🔍 Search keywords (e.g. OpenAI, NUST, Admissions)... ] [⚡ All Sources] [Run] |
+-----+
| [📊 Sentiment Analytics] [📈 Topic Breakdown] [❤️ Likes] [💬 Comments] [🔄 Shares] |
+-----+
| [All Feeds] [X] [Facebook] [Reddit] [YouTube] [Instagram] [LinkedIn] [News] |
|
| POST CARD:
| Author: @OpenAI • Platform: YouTube • Sentiment: Positive
| Text: "OpenAI Sora AI Video Demo: Tokyo Walk in Snow #shorts #sora"
| [🎬 Video Player Preview with Play Badge ]
| [🎞️ Verify Video ] --> [📺 LIKELY AI VIDEO (SORA) (98%) ] [ Verified ✓ ]
| [👉 Text: Human (80%) ] [📰 Fact-Check: SUPPORTED ] [ Citations: 4 Sources ]
+-----+
```

4.1 Running Multi-Platform Searches

- Search Input:** Enter target query terms (e.g., `OpenAI` , `NUST` , `Admissions 2026`) in the primary search bar.
- Platform Selector:** Select `⚡ All 7 Sources (Parallel)` or isolate a specific network.
- Retrieval Limit:** Select between 10, 20, 50, or 100 posts per query.
- Execute:** Click `Run Scan` or press `Enter` . The system queries all selected collectors concurrently and displays normalized results chronologically.

4.2 Using the Advanced Boolean Query Builder

Click the query syntax chips or open the builder modal to compose complex search logic:

- Boolean Grouping:** `AND` / `OR` logic blocks.
- Exact Matches:** Quoted phrases (e.g., `"artificial intelligence"`).
- Account Filters:** Target specific creators (e.g., `from:OpenAI`).
- Media Filters:** Isolate rich posts (e.g., `filter:media` , `filter:links`).
- Engagement Thresholds:** Set floors (e.g., `min_faves:100` , `min_retweets:20`).

4.3 Authenticity & Fact-Checking Triage

Every post card displays an integrated authenticity bar:

- **Text Analysis Badge:** Indicates lexical entropy evaluation (**Text: Human (80%)** vs **Text: AI-Gen (85%)**).
- **Fact-Check Status:**
 - **SUPPORTED** Corroborated by independent news wires. Click **Extracted Claims & News Citations** to inspect cited sources.
 - **UNVERIFIED** Factual assertion made, but no independent accredited publisher has verified it.
 - **NO CLAIMS** Casual commentary or opinion without empirical factual claims.

4.4 On-Demand Image Forensics

1. Locate any post card containing an image.
2. Click **[✨ Verify Image]**.
3. In < 3 seconds, the system computes the 2D FFT spectral peak density and Gaussian noise kurtosis directly in RAM, displaying either:
 - **LIKELY AI GENERATED** (with kurtosis values > 14.0 and lattice densities).
 - **AUTHENTIC CAMERA PHOTO** (confirming natural sensor PRNU profile).

4.5 On-Demand Video Forensics (YouTube Shorts & Instagram Reels)

1. Click the **YouTube** or **Instagram** platform tab.
2. Locate the video card with the embedded thumbnail and play overlay.
3. Click **[🎬 Verify Video]**.
4. The system samples 4 keyframes concurrently into RAM, applies Gunnar Farneback dense optical flow tracking and PRNU noise kurtosis, displaying:
 - **LIKELY AI VIDEO (SORA / KLING)** (detecting non-rigid warping and synthetic kurtosis).
 - **NATURAL CAMERA VIDEO** (confirming rigid real-world motion and physical sensor profile).

5. Platform Account & Session Management

For channels requiring authenticated sessions (e.g., protected X/Twitter feeds or private Facebook groups):

1. Click **⚙️ Scraper Settings** in the top navigation bar.
2. Select the target platform (**X** , **Facebook**).
3. Paste session cookies in JSON or Netscape format.
4. Click **Save & Register Account**. TrendChecker encrypts and stores the cookies in the local database for seamless rotation.
5. If rate-limits are encountered, click **Reset Rate Limit Locks** to cycle accounts immediately.

6. Troubleshooting & Operational FAQs

Q1: Port 8000 is already in use by another local service.

Resolution: Start the server on an alternative port by passing the `--port` flag:

```
python -m uvicorn app.web.app:app --host 127.0.0.1 --port 8080
```

Then access the dashboard at `http://127.0.0.1:8080`.

Q2: ModuleNotFoundError: No module named 'cv2' or 'numpy'.

Resolution: Ensure your virtual environment is active and reinstall the computer vision stack:

```
pip install opencv-python numpy scipy pillow
```

Q3: Optical flow calculation returns NATURAL CAMERA VIDEO on an AI video.

Resolution: Ensure the video is not a heavily re-compressed screen recording. Highly compressed media (< 240p) blurs high-frequency noise residuals. The system automatically weights contextual disclosures (e.g. `#sora` , `#kling`) more heavily in low-resolution conditions.

7. Operational Health Checklist

Pre-Deployment / Evaluation Verification Checklist

- ✓ **Unit Tests:** Run `python -m pytest tests/` → Confirm 38/38 passing tests.
- ✓ **Server Launch:** Run `python run_web.py` → Confirm server initializes on `http://127.0.0.1:8000` .
- ✓ **Multi-Channel Scan:** Execute a test query (e.g., `OpenAI`) → Confirm 7-platform feeds load chronologically.
- ✓ **Forensic Verification:** Verify an AI Short and an authentic Camera Short → Confirm purple and green triage badges render dynamically.